

Notes

Wednesday, May 1, 2024 11:03

Some background

Euler number or Euler zigzag numbers

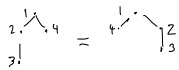
 $A_n: 1, 1, 1, 2, 5, 14, 61, 292, \dots$

$$2A_{n+1} = \sum_{k=0}^n \binom{n}{k} A_k A_{n-k}$$

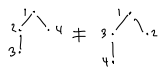
Cases of a lot.

increasing (11)-tree:

promising nodes are so that every non-leaf node has two or three children. Thus, if the tree has n nodes, we can consider the tree as a rooted tree. In fact, a way to reach the root node. Thus, we marked all the nodes as root. For example



But


 $f(n) = \#$ of different ways to cross trees of n vertices

Lemma 1.

$$\sum_{n \geq 0} \frac{x^n}{n!} = \sec x + \tan x$$

Proposition: $f(n) = E_n$ $n=1$ $f(1)=1$ $n=2$ $f(2)=1$ $n=3$ $f(3)=2$ $n=4$ $f(4)=5$ Proof: Let $y = \sum_{n \geq 1} f(n) \frac{x^n}{n!} = x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$ Take the derivative: (w/ respect to x)

$$y' = \sum_{n \geq 1} f(n) \frac{x^{n-1}}{(n-1)!} = \sum_{n \geq 0} f(n+1) \frac{x^n}{n!} = y$$

Let's look at different cases for trees of n nodesa) A single vertex: ($n=0$)b) Has one vertex of the root which is an increasing (1,2)-tree with n vertices

$n=3$ $\leftarrow$ already counted from $\leftarrow$ the ok y possible trees for ($n=3$) vertex or pos (perm, permutation for 1, 2, 3)

c) Has two vertices of the root, each of which is an increasing (11)-tree with n vertices in all.

$n=3$ This will have to do with possible for ($n=3$) vertex or pos (perm, permutation for 1, 2, 3)

From this observation, we get

$$y' = 1 + y + \frac{1}{2} y^2 \rightarrow \text{Check and if you can't get it, not given to get.}$$

This has a unique solution: $y = \sec x + \tan x$

Why?

$$y' = \sec x \tan x + \sec^2 x$$

$$y' = 1 + y + \frac{1}{2} y^2 = 1 + (\sec x \tan x + 1) + \frac{1}{2} (\sec x \tan x + 1)^2$$

$$\begin{aligned} y' &= \sec x \tan x + \frac{1}{2} \sec^2 x + \sec x \tan x + \sec x \tan x + \frac{1}{2} \tan^2 x + \frac{1}{2} \\ &= \frac{1}{2} (\sec^2 x + 2 \sec x \tan x + \tan^2 x + 1) \\ &= \sec x \tan x + \frac{1}{2} (\sec^2 x + \tan^2 x + 1) \\ &= \sec x \tan x + \frac{1}{2} \left(\frac{1}{\cos^2 x} + \frac{\sin^2 x}{\cos^2 x} + \frac{\cos^2 x}{\cos^2 x} \right) \\ &= \sec x \tan x + \sec^2 x \end{aligned}$$

This is what we wanted.

So we have

$$y = \sec x + \tan x = \sum_{n \geq 1} f(n) \frac{x^n}{n!}$$

$$\left(\sum_{n \geq 0} E_n \frac{x^n}{n!} \right) \text{ from lemma 1}$$

$$\text{So } f(n) = E_n \text{ (taking from } E_1)$$